From the Anthropocene to AIGW: The Narrative Autonomy of Generative Imagery

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Abstract

The rise of generative artificial intelligence is shifting image production from camera- and film-centered recording toward worldbuilding continuously generated through data, models, and platform workflows. Taking AI-generated worlds (AIGW) as its central object, this article first revisits how traditional cinema presents a world picture through the media practices of viewing and archiving within the Anthropocene. It then introduces planetarity as a theoretical perspective for recalibrating anthropocentric scales and illusions of governance. The analysis proceeds through three dimensions: planetary vision, planetary tactility, and planetary thinking. At the visual level, it reveals AIGW's de-geographical organization of space and diffusion-based recombination of data. At the tactile level, it examines how the interpenetration of boundaries and digital embodiment transform the subject from a spectator into a situated participant drawn into a generative environment. At the level of thinking, it follows technological, noetic, and ecological diversity to explain how AIGW's world generation establishes its planetary conditions between rules and consequences. Finally, the article argues that the narrative autonomy of generative worlds is realized in a planetary sense through three levels: generative autonomy, causal autonomy, and normative autonomy. It thereby outlines a field of inquiry into image ethics attentive to infrastructural constraints and planetary costs.

Keywords: Anthropocene; planetarity; AIGW; generative imagery; artificial intelligence; narrative autonomy

Introduction

In recent years, the rapid development of artificial intelligence has brought image-based narrative into a new technological context through generative technologies. Since 2024, the large integrated gaming platform Roblox has progressively opened access to generative AI creation tools designed to accelerate game-asset production. In November 2025, Roblox announced early access to in-experience “4D functional objects” on its developer forum: players and creators could use text during a running experience to generate objects with gameplay functions. Static game assets acquired interactive, adaptive intelligence, becoming infrastructural and constitutive elements of virtual game worlds. Similarly, since 2025, mainstream game companies, led by Ubisoft, have deployed AI non-player characters (NPCs) in their game design. Able to converse and collaborate, these characters participate in interactive decisions and in the development of the game world. At both visual and narrative levels, the metaverse imagined in *Ready Player One* is gradually taking preliminary form through generative AI. The imagined digital worlds of cyberpunk are shifting from audiovisual spectacle into objects that can be comprehended.

In reinforcement learning and embodied-intelligence research, a “world model” generally denotes a model that simulates environmental dynamics. Given a current observation and action, it predicts the next observation and longer-term developments. In its 2024 paper “Genie: Generative Interactive Environments,” DeepMind introduced Genie as a generative interactive environment trained without supervision on unlabeled internet videos. It can generate action-controllable virtual worlds from text,

synthetic images, photographs, and even sketches. The study proposes that, at a scale of 11 billion parameters, Genie can be considered a "foundation world model."[1] Although Genie currently remains principally a research and prototyping tool rather than a widely adopted mass-market product, AI technologies can already be seen gradually realizing real-time generative world models through the construction of three-dimensional imagery with physical structure and memory.

Digital imagery driven by generative technologies is thus gradually transforming its status as a narrative object, forcing worldbuilding, a classic issue in film studies, into a new technological and epistemological field. Moving away from traditional cinema's pursuit of the "myth of total cinema" in André Bazin's account and its recreation of the illusion of the real world, imagery no longer depends primarily on the camera's indexical recording of an existing material reality. Increasingly, it emerges through generation and diffusion across data, models, and platform workflows. Authority over worldbuilding shifts from a traditional production chain composed of human authors, scripts, and mise-en-scène toward a self-organizing tendency learned in advance from data and systemic conditions.

This article calls image-worlds continuously produced and maintained in a consistent state by generative systems "AIGW" (AI-Generated Worlds), denoting models' generation and simulation of interactive environments. The concept was first proposed in May 2023, in an article published on Medium by the Dreamix intelligent-systems team, identified in the source text as a Google team, entitled "AIGC Is an Inflection Point that Will See the Emergence of AIGW." It announces a movement toward full-stack AI-generated worlds and treats the rapid adoption of AIGC as a creative tool as the key technological advance enabling their emergence.[2] An awareness of AIGW appeared at the beginnings of generative AI creation and has remained the ultimate imaginary of AIGC development. We need to recognize the revolutionary changes that the emergence of this medium brings to the arts. Like the invention of photography in the nineteenth century, it may reshape the future industrial systems of film, television, and game production, as well as how worlds are constructed and understood.

This conception invites associations with, and discussion of, the "metaverse." The science-fiction novel *Snow Crash* first described this vision: human beings live through digital avatars in a virtual digital environment constituted by sensory immersion, a digital economy, and communal dependence. Originating in the real world while running parallel to it, this environment entails a symbiosis of the virtual and the real. As communication technologies and computing power advance, AI redirects the spectacular conception of the metaverse, focused on spatial and institutional frameworks, toward a mechanism of worldmaking predicated on models: AIGW.

In contrast to this generative imaginary, established cinematic practices use photographic media to faithfully record ecological transformations and ideological changes involving the Earth, society, and nature, rendering landscapes of "manufactured nature" visible. "If any medium is especially adept at capturing manufactured nature, it is cinema."[3] Within traditional cinema, human beings are the subjects of worldbuilding and planetary transformation, while imagery performs the audiovisual translation of human intentions. In Bazin's metaphor, cinema preserves, like a mummy, humanity's creation and destruction of the world, allowing us to understand the relationships between human beings and nature, technology, modernization, and the Earth as a whole. This is the Anthropocene story cinema tells, and the Anthropocene perspective through which we have hitherto viewed the world.

The Anthropocene fundamentally connects the human species with the Earth, prompting reflection on human actions and their effects at a planetary level. Today, this perspective calls for a renewed examination of AIGW through planetarity. Planetarity is a new direction in film studies that moves beyond Western-centrism and anthropocentrism, expanding and deepening our understanding of the world through time, space, and the Earth's materiality. Confronting the Anthropocene crisis in an age of technological

singularity, a planetary framework of understanding can clarify the ontology of cinema and other digital media.

The logic through which imagery constructs worlds is undergoing a shift from anthropocentrism toward a planetary logic. Within the Anthropocene, imagery, as a material medium possessing geological force, turns humanity's transformation of the world into an "image fossil." Yet when generative AI shifts the infrastructure of film production from traditional human collaboration toward planetary-scale reconfiguration, image generation moves from individual, indexical production to synthetic generation based on algorithmic neural networks. This ontological rupture signals a relinquishment of human subjectivity within image sovereignty and opens new possibilities for narrative autonomy under the logic of nonhuman intelligence.

On this basis, and through the planetarity of cinema, this article examines the following questions: How do traditional cinema and generative imagery in the AIGC era differ in their ways of presenting the world? As human subjectivity in image creation gradually dissolves in the age of generative technologies, how can intelligent imagery construct AIGW through the planetary reconfiguration of media, depart from anthropocentric narrative paths, and achieve narrative autonomy and logical evolution in a substantive sense?

I. The World Picture of Traditional Imagery in the Anthropocene

In *Koyaanisqatsi* (1982), Godfrey Reggio presents a montage of striking accelerated images: vast mining operations, pulsing urban telecommunications, and circulating traffic. Rapidly moving crowds, laboring bodies reshaped by assembly lines, nightscapes taken over by lighting and advertising screens, and spaces connected by transport and energy systems form an almost automatically operating landscape of manufactured nature. Although the film's narrative themes are ecological crisis and environmental problems, it expresses a deeper central symptom of human society: the more the world is transformed into something calculable and administrable, the more individual experience is caught within irreversible systemic acceleration. Since the 1950s, the effects of human activity on nature have become increasingly evident, most directly in phenomena such as frequent acid rain, the greenhouse effect, and deteriorating air quality. In May 2019, following a vote, the Anthropocene Working Group proposed "Anthropocene" as a scientific term describing a new iteration of the Earth's ecological environment, identifying the mid-twentieth century as its beginning.[4] The concept marks an era in which human activities comprehensively affect the Earth's environment, pointing to the decline of human civilization and its systems of intellectual and cultural expression and the birth of a civilization governed by technology.

If Bazin's mummy complex concerns the human body's anxiety about resisting time, Anthropocene imagery concerns the traces humanity leaves in the natural world as a geological force. In documentaries about cities and nature such as *Koyaanisqatsi*, we see humanity's explosive productivity and creativity, but also sense an unsettling destructive power: human forces are quietly reshaping the relationship between people and the Earth. As a human artifact and a medium inseparable from scientific and technological development over more than a century, cinema's narrative functions inherently reflect upon and critique the external environment, modernity, capitalism, and anthropocentrism. They have a distinctly Anthropocene orientation. To understand imagery's position within the Anthropocene, we must release it from purely aesthetic representation and return to the medium's material properties. Drawing on media archaeology, Sun Hongyan proposes placing cinema's materiality within the category of "Anthropocene fossils" or "Anthropocene objects." She argues that "cinema as archive" in the Anthropocene can be understood as an "image fossil." Instincts rooted in humanity's biological inheritance impel people to use cinema to record their species' civilization and material memory, together with shifts in power relations between humanity

and the Earth, forming an archival recording apparatus. This property of cinema also responds, at the level of material ontology, to Latour's proposition that, in the Anthropocene, “the world turns toward the standpoint of things.”[5]

Reexamining imagery's logic of world construction through the Anthropocene first requires a return to Heidegger's classic ontological proposition concerning “the age of the world picture.”[6] Within traditional media, cinema has consistently retained the function of presenting objective things as visual signs. Through their indexical relationship with reality, photographic media turn surface changes and material consequences that are otherwise dispersed, slow, and difficult to grasp into images that can be viewed, compared, and archived: extractive landscapes, industrial ruins, immense infrastructures, accumulations of pollutants and waste, and unusually frequent climatic disasters. These represented natural landscapes manifest modern technological and capitalist logics within social development. Cinema considered in the Anthropocene therefore need not depend on explicit ecological themes. Its deeper function is to return transformations of the world from abstract accounts to visual form. Through viewing, we recognize that the world no longer corresponds to its former conception as a natural whole but has become a composite system continuously produced and administered. Cinema organizes scars on the Earth's surface, resource flows, the incorporation of laboring bodies, and the rhythms of urban life within a single structure, transforming the Anthropocene's overarching contradictions into a visible world picture.

The “image fossil” also has a second, material meaning here. Cinema's own medial properties become part of the world picture, participating as material products of modern civilization in the world's generation and consumption. In *Finite Media: Environmental Implications of Digital Technologies* (2017), Sean Cubitt treats media as ecological artifacts. Discussing the restoration of a damaged feature film, he explains that celluloid, composed principally of cellulose nitrate, volatilizes and decomposes over time, eventually becoming sticky, deformed, and unprojectable. Cinema's archival struggle against entropy can be understood as a decay of life. Digital film restoration, as collaboration between chemical processes and digital intelligence, symbolizes how humanity, technology, and nature mediate one another: amid continual, catastrophe-like dissolution and flaking, the reorganization of frame-sequence information temporarily sustains the image.[7] Imagery therefore depends on materials, energy, infrastructure, and global supply chains. Film's chemistry, the storage and computation of digital images, the energy consumed in projection and transmission, and hardware replacement all embed cinema within the Anthropocene's material cycles. Understood as an Anthropocene archive, cinema preserves human creation and destruction of the world; it may also itself become an “image fossil” in future strata, remaining within planetary history as an “Anthropocene object.” The worlds constructed by traditional media carry material costs and political-economic structures. Which landscapes are filmed, which disasters become visible, and which forms of labor and resources are obscured constitute an ideological cross-section of the Anthropocene world picture.

In this sense, image landscapes exhibit an internal tension. They require a totalizing perspective to grasp crises at a planetary scale, yet must guard against the illusion such a perspective induces: that, once sufficiently represented and panoramically viewed, the world can be calculated and governed. This article therefore takes planetarity as a means of recalibrating and comparing the Anthropocene world picture. The Anthropocene's material controllability must be distinguished from the planetary age's alterity, so that worldbuilding no longer presupposes mastery but returns to this planet's inexhaustible material processes, deep time, and structures of responsibility. By this criterion, AIGW's emergence becomes decisive. When image-worlds cease to rely primarily on photographic traces for recording and representation and turn instead toward worldmaking continuously generated and held consistent through data, models, and platform workflows, the testimonial logic, archival logic, and organization of scale underlying traditional cinema's Anthropocene world picture are rewritten.

II. Generative Image-Worldbuilding within a Planetary Genealogy

Some years ago, the digital game *No Man's Sky* established an open universe driven by deterministic procedural generation, theoretically containing a combinatorial space of approximately 2^64 planets. Players explore and survive in a "nearly infinite" galaxy: jumping between systems, landing on planets, scanning flora, fauna, and minerals, collecting resources, trading, and fighting. They can upload "discoveries" to a shared database so that other players encounter the same named and recorded traces of a world at the same coordinates. In this game, the "planet" is infinite and open, yet can be generated, determined, and reproduced.

As early as the 1950s, the Greek-French thinker Kostas Axelos established, in his lectures, a philosophical concept of the "planetary" combining the dimensions of "world" and "play." "World" describes how the whole worlds itself into an experiential totality: a difficult-to-name but ever-present "it." "Play" explains how the world moves and unfolds through time and becoming - its mode of movement. The planetary, in turn, thinks world and play as a decentered total field. It renders categories inoperative, allows questions to continue emerging through wandering, and draws human beings into this field rather than placing them at its center.[8]

The subsequent popularity of this concept has been seen as offering a possible response to the technological singularity and the Anthropocene crisis. Later studies of planetary theory entered film and media studies. In his research on "planetary cinema," Han Xiaoqiang argues that, from such a position, we should retain existing technological conditions while extending our concern beyond humanity or life to the "earthbound" in a planetary sense: humans, rocks, plants, and cinema all belong to it. He clarifies the film medium's planetarity through a geographical logic moving from surface to geology and then to planet, and grasps planetary cinema and its implications through planetary vision, planetary tactility, and planetary thinking.[9]

When image production enters the context of AIGW, this triad becomes a methodology for analyzing how generative worlds are constructed and sustain themselves. The visual experiences generated by world models, the tactile structures constituted by interactive feedback and infrastructural material chains, and the planetary thinking that coordinates scale, agency, and ethico-political questions together form the conditions for generative imagery's movement toward narrative autonomy.

1. Planetary Vision: De-geographicalization and Data Recombination

The German naturalist Alexander von Humboldt was an early explorer of planetary media. His concrete practices concerning the Earth and images, including the *Naturgemälde*, laid foundations for early Earth-system thinking and global comparative methods. He inaugurated a tradition of knowledge that made the Earth legible, providing a frame of reference for the totalizing landscapes of photography and cinema and even today's modeled worlds. This constituted his visual exploration at the level of planetary geography.[10] If planetarity initially enters image practices through an impulse to rescale, demanding an understanding of the world's overall relations across larger spatial dimensions, "planetary vision" communicates the world picture produced through a medium's capacities. Through comprehensive views, totalization, and navigable viewing positions, imagery organizes surface changes, infrastructural networks, and environmental consequences otherwise dispersed across local experience into detailed and comparable visual wholes.

If the totalizing gaze of early planetary media presupposed geographically traceable visual technologies such as aerial photography, remote sensing, and cartographic systems, it always carried, in intellectual history, the shadow of an "Archimedean perspective": a view that objectifies the Earth. Hannah Arendt's critique of space imagery suggests that the 1960s Archimedean perspective, presenting the Earth from

space, is structurally homologous to a worldview of technological imperialism. It can be understood as a concrete manifestation of Heidegger's "age of the world picture": the world seemingly becomes eligible to be known only after being placed within an overarching framework, a measurable scale, and a comparable coordinate system.[11]

The emergence of generative imagery and AIGW, however, produces a structural displacement of this totalizing gaze: planetary vision is generated through a de-geographical visual regime. De-geographicalization here means more than the image's release from physical presence at a filming location. It also means that the generation of world pictures increasingly depends on a "globe in the computer." In *Death of a Discipline*, the Indian-American literary theorist Gayatri Chakravorty Spivak proposes that the planet overwrite the globe: the globe's presence in our computers induces a powerful impulse to control it, whereas the planet signifies an alterity that cannot be fully possessed. We inhabit it only on loan.[12] In the information age of AIGC worldbuilding, when the world as a data object is incorporated into models, platforms, and computing infrastructures, planetary vision becomes more susceptible to the illusion of governance implied by the computerized globe. At the same time, planetarity requires us continually to recall that modeling does not exhaust the world: it retains a remainder and an alterity beyond the model. AIGW thus exhibits two intertwined sets of characteristics at the level of planetary vision.

The first is the movement from geographical to platform scales: de-geographicalization at the spatial level. Traditional planetary vision still requires material conditions such as geographical systems, mapping institutions, and infrastructural networks to secure visibility. In generative imagery, planetary vision more closely resembles a platformized panorama. It no longer begins at a camera position in a particular place, but takes training data, the allocation of computing resources, model architectures, and interface permissions as new points of access. This change in how world pictures are organized also transfers visual power from human control of media to the governing logics of models and platforms.

The second is the model's dimensional reduction and recombination: diffusion-based transformation at the data level. The world pictures of generative imagery depend increasingly on statistical reconstruction. Multimodal image records of a three-dimensional world are vectorized and recombined at the data level through noising and denoising before returning to the screen in perceptible form. Vision departs from the guarantee of truth furnished by material presence, adopting planetary distribution, probability, and interpolation as its generative mechanisms. The world is reduced to binary data and reconstructed through diffusion into visual images. Planetary vision thereby acquires a "nonhuman field of vision," exceeding the limits of the human eye and a single camera position. It depends not on localized bodily experience but on cross-scale data integration, pattern recognition, and generative inference. OpenAI described Sora, its video model released in 2024, as a "world simulator." When video-generation models are trained on large-scale data, they exhibit capabilities that seem to simulate the world, particularly spatial and temporal consistency, causal traces, and approximate environments. OpenAI's technical report states this explicitly in its title and introduction, presenting the scaling of video-generation models as a promising path toward general-purpose simulators of the physical world.[13]

2. Planetary Tactility: Digital Embodiment and Human-Machine Symbiosis

To discuss tactility is to address the relationship between media and bodies. Planetary tactility requires us to understand “touch” as a medial-material process rather than simply a human sensory act. Planetary cinema theory has examined geological filmmaking, which reconnects imagery with geochemistry and deep time through film's material substrates and production chains.[14] In her research, experimental filmmaker Sasha Litvintseva explores “contact” in terms of external boundaries and their transgression, and the perceptible and imperceptible. Her film *Asbestos* (2016) follows asbestos, a material between visibility and invisibility, to investigate the difference between metaphorical touch and actual physical contact. It thereby develops a cross-scale perceptual framework, tracing entanglements from molecule to planet and from present time to deep time.[15] Whereas traditional cinema's materiality can remain concealed behind image appearances, generative imagery is attempting to reassert the necessity of returning to media materiality. The computing resources, data centers, hardware, and energy systems on which AI depends attest to planetary-scale material circuits and geological politics.

Tactility in AIGW does not mean more realistic haptic sensations. It appears first as the interpenetration of boundaries. The clear divisions that screens and interfaces establish between subject and object, viewer and viewed world, gradually dissolve. Imagery becomes a membrane that must continually be traversed and endured. Elizabeth A. Povinelli likens an organism's separation from its exterior to a “final membrane,” while planetary geology seeks progressively to expand that membrane's scope in order to perceive the respiratory tension between the biological and the geological.[16] Subsequent planetary cinema research holds that geological cinema's excavation of the membrane necessarily breaks through the surface, forming a “hole.”[17] The hole's emergence undoubtedly reshapes planetary affinities between life and nonlife. In interactive generative worlds, the gamified scenes, real-time rendered spaces, and continually updated image environments we experience immersively transform viewers of traditional imagery into inhabitants. The subject no longer stands outside the world inspecting its appearances, but is shaped by it through continual response and feedback. Planetary tactility thus becomes a critical-zone-like media experience, emphasizing fluid boundaries, unstable subject positions, and the mutual embedding of humans and environments such as large models and digital platforms.

AIGW's second tactile characteristic is digital embodiment. Unlike mechanically mimicking the hardness or softness of the real world, embodiment emphasizes a sense of body ownership and sensory involvement. The well-known “rubber hand illusion” in cognitive science shows that artificial body parts can be experienced as part of the self. Generative digital situations with continuous feedback can trigger cross-sensory associations between the physical and virtual body, producing an illusion of body ownership.[18] At GDC 2024, Inworld and NVIDIA collaborated on a technology demonstration called Covert Protocol, showcasing NVIDIA ACE and the Inworld Engine. Through speech recognition and facial animation, among other techniques, the demo enables immediate AI recognition of, and feedback on, tones, pauses, and expressions in exchanges between humans and digital environments. Through continuous responses, participants acquire involvement and presence. Through controllable bodily proxies, responsive environmental feedback, and continually generated situational coherence, AIGW gives subjects an experiential tactile logic of being drawn into the world. Continuous feedback draws the subject into the digital world's own critical zone, returning it from a distant, external viewing position to an interface of conditions composed of dependence, friction, and consequences. In this sense, digital embodiment constitutes a planetary dimension of AIGW. Rather than intensifying mastery over the world, it situates the subject in a critical relationship with the environment. Worldbuilding becomes a structure of existence shared with nonhuman processes.

Planetary tactility also entails the hybridization of human life and nonlife. Beyond their role as external instrumental objects, generative systems and intelligent agents become symbionts within generative image environments. Benjamin H. Bratton conceives of the Stack as a planetary-scale infrastructure encompassing forms of planetary computation such as intelligent networks, cloud computing, and mobile software.[19] Furthermore, Kate Crawford emphasizes in *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence* that the stack required by AI systems is far larger than hardware technologies such as servers and networks. AI's core can even be described in terms of "planetary minerals," exemplified by lithium, embedded within a full-stack supply chain of capital, labor, and resources.[20] AIGW's "tactility" thus traverses an entire metabolic chain, from mineral extraction to computing consumption, and from data governance to platform orchestration. Planetary tactility consequently leads inevitably toward an ethical dimension of mutual care involving humans. At the threshold between the waning Anthropocene and the beginning of an AI age, humanity is assigned an important task: to establish a positive biopolitical relationship with nonhuman intelligence, making possible a transition from artificial to planetary intelligence.

3. Planetary Thinking: Technology, Mind, and Ecology

Technological acceleration drives globalization's seemingly inexorable promotion of capital and material flows. The accelerating system it constructs is changing the course of human civilization and the human world as a whole. If philosophy is indeed coming to an end, as Martin Heidegger suggests, our task remains to reflect on its nature and future and seek another beginning. Avoiding the polarization between celebrating technological domination and demonizing technology, planetarization is an inevitable tendency in the redirection of history: the total mobilization of matter and energy across geographical and psychic dimensions, opening the possibility of technology. On this basis, the philosopher of technology Yuk Hui proposes "planetary thinking" in "For a Planetary Thinking": a new philosophical framework to replace "global thinking." Hui describes its internal structure through three diversities. Biodiversity concerns relations between humans and nonhumans inscribed and mediated through technical inventions and embedded in the life of a people. Noodiversity critiques the noosphere of Pierre Teilhard de Chardin and its vision of a global technological collectivity converging into a superbrain. Instead, fragmented, diversified forms of mind, combined with technodiversity, should reconfigure nonhuman institutional relations and political economy. Technodiversity is the precondition and priority of the other two: homogenization cannot serve as a universal standard, and the myth of technological neutrality inevitably entails subjection to development.[21]

AIGW is based on models' ability to generate images, scenes, and interactions. In this sense, such an ability should generate and sustain differences under planetary conditions, while making their consequences bearable and accountable. Technodiversity, noodiversity, and biodiversity together constitute the conditions of AIGW's planetary existence.

Hui's technodiversity concerns the possibility that understandings, constructions, and value orientations of technology as technology may develop differently across localities and worldviews. Otherwise, planetarization merely flattens difference into a single infrastructural order. Generative image-worlds possess mechanisms capable of forking. The question is whether different generative rules, forms of play, and narrative structures can coexist rather than be assimilated through global replication. An extreme example of a single generative grammar is *Microsoft Flight Simulator*. Based on Bing Maps' enormous datasets, AI-assisted procedural processes generate vast quantities of terrain features across the globe. Countless dispersed rendered objects, each with its own operational logic, jointly construct a digital planetary world.[22] This underscores that technodiversity in the sense of planetary thinking allows rules, data, operational locations, and interactions to branch, so that worlds need not always be produced in the

same technological form.

Noodiversity takes the generation of worlds further, toward the generation of subjects. It critiques the traditional convergence of mind, opposing the unification of cognition, attention, and judgment within one technological rationality and one structure of experience. If the narratives of generative imagery treat generation merely as storytelling in place of humans, they remain at the level of content. When they alter subjects' attention and actions through feedback regulation and stochastic diffusion, however, they correspond to noetic freedom at a planetary level. “Generative Agents: Interactive Simulacra of Human Behavior” (Joon Sung Park et al., 2023) proposes AI agents with a closed loop of memory, reflection, and planning. In a small-town-like sandbox environment, the agents spontaneously form social interactions and chains of events; users can also interact with them through natural language.[23] AIGW's creative capacity lies principally in the joint emergence of multi-agent inputs and environmental feedback. Subjects themselves move from interpreting texts to judging and managing generative consequences, while resisting platformized thinking that trains every experience into a single mental pattern.

Without biodiversity, AIGW's planetarity also becomes an illusion: it produces a de-geographical totality at the visual and interactive levels while treating energy consumption, materials, supply chains, and ecological consequences as invisible presuppositions. Planetary thinking requires worldbuilding to be grounded anew in planetary metabolism and to consider and assume responsibility for ecological consequences. In Atlas of AI and in Anatomy of an AI System, her collaborative project with researcher Vladan Joler, Kate Crawford reveals AI's planetary costs: it is an extractive system composed of mineral resources, energy, and human labor, rather than pure intelligence in the cloud.[24] At the level of verifiable data, the International Energy Agency reports that data centers consumed approximately 415 TWh of electricity in 2024, around 1.5 percent of global electricity consumption, with significant growth in recent years.[25] Biodiversity therefore requires AIGW to assume the material consequences of its infrastructure and incorporate them into the conditions it considers internally, enabling more binding governance relations between humans and nonhumans. Only then can AIGW move from planetarization's total mobilization toward the generation of diversity in the sense of planetary thinking.

III. The Narrative Autonomy of AIGW

Imagery in the Anthropocene records and interprets representations of a world picture. AIGW, constructed through generative imagery, instead takes models, platforms, and interaction as the technological basis for endlessly emergent production. Reducing its narrative generation to a question of creative authorship shifting from human to machine is too narrow. We should attend instead to how, within this mechanism of creation and beyond storytelling by human intelligence, such a world landscape reconfigures human-nonhuman relations in a planetary sense and achieves a more natural experience. This directs attention to how AIGW can achieve narrative autonomy.

First, narrative autonomy manifests itself at the generative level. The narrative text moves from closure to openness to shape a world environment capable of continual emergence and persistence. In other words, AIGW's narrative starting point is not plot. It can generate new events while continuing prior states and styles. Chen Xuguang has explored the ideal of a “world model” for generative imagery: systems such as Sora are understood as the ultimate imaginary of a path toward world models, generating through prediction “new matter that has never existed in history.” WHAM (World and Human Action Model)/Muse, developed through collaboration between Microsoft Research and Ninja Theory, approaches the imaginary of AIGW. The model continually updates gameplay sequences, maintaining consistency and the persistence of modifications to some degree, thereby supporting gameplay ideation and iterative prototyping. Generative autonomy thus lies in a world-operating mechanism realized through the

computation of layered scenes.

Second, narrative autonomy manifests itself at the causal level, requiring further inquiry into how a world answers for actions. Causality in traditional image-based narrative depends more on viewers' interpretation and supplementation of meaning. In AIGW, for narrative to become a system of continuous governance, chains of consequences must enter its structure, enabling panoramic visualization at a planetary level. GameNGen, presented in “Diffusion Models Are Real-Time Game Engines,” offers an instructive extreme case: researchers use a diffusion model for next-frame prediction to simulate the classic game Doom across long interactive trajectories while preserving playability at real-time frame rates. This requires the model's causal plausibility to extend beyond the visual level to the steps and logic of generation itself.

Finally, the narrative coherence of generative imagery should rest on normative autonomy. AIGW generates scenes and events alongside norms and principles that run throughout the world's unfolding. For autonomy to hold in a planetary sense, it cannot mean an infinitely expanding generative capacity. It must instead establish a structure incorporating boundaries and thresholds: which generations and actions are permitted, at what intensity, how transgressions are blocked or corrected, how harms are repaired, and how responsibility is distributed among subjects, agents, and systems. Only when norms enter interaction and narrative causality as internal world variables, rather than invisible adjudications at a platform's back end, can AIGW's narrative autonomy achieve its ethical realization in a planetary sense.

Conclusion

The future orientation of generative imagery toward AIGW shifts the construction of world pictures beyond aesthetic and narrative debate toward reflection on technological infrastructure and mechanisms of production. Imagery no longer primarily aims at representation; under conditions set by models, platforms, and computing resources, it intervenes in how worlds are organized and experience is structured. The resulting challenge is how established frameworks of media criticism can respond to a generation of reality predicated on data, computation, and interaction as the boundaries of creative subjectivity blur. Future research must advance on three levels. First, it must redefine the authenticity and historicity of images and examine the conditions under which image-based facts become established within generative systems. Second, it must situate AIGW within planetary-scale political-economic and environmental conditions, specifying how resource consumption, labor organization, and governance structures internally shape image forms, while avoiding the equation of technological progress with narrative emancipation. Third, it must strengthen interdisciplinary approaches by combining film studies' formal analysis with the normative discussions of the philosophy of technology to develop deeper criteria for evaluation. Only when these questions have been fully explored can generative imagery be understood as a media practice possessing historical consciousness and responsibility toward reality, rather than merely a new industrial tool or cultural landscape.

Notes and References

The original's 25 numbered notes are retained. Established English names and work titles have been standardized. Chinese-language sources are identified as such, and their article titles are translated. Quotations not verified in their original editions are translated from the supplied Chinese text.

[1] Jake Bruce, Michael Dennis, Ashley Edwards, et al. “Genie: Generative Interactive Environments.” arXiv, February 23, 2024. https://arxiv.org/abs/2402.15391

[2] Dreamix. “AIGC Is an Inflection Point that Will See the Emergence of AIGW: We're Headed Towards Full Stack AI Generated Worlds.” Medium, May 25, 2023. https://medium.com/@DreamixOfficial/aigc-is-an-inflection-point-that-will-see-the-emergence-of-aigw-4d9bb2da1475

[3] Jennifer Peterson and Graig Uhlin. “In Focus: Film and Media Studies in the Anthropocene.” Journal of Cinema and Media Studies 58, no. 2 (2019): 142-146.

[4] Wang Youran. “‘The Anthropocene’ Highlights the Importance of the Humanities and Social Sciences.” Chinese Social Sciences Today, November 15, 2021. In Chinese.

[5] Sun Hongyan. “Contemporary Western Film Studies from an Anthropocene Perspective.” Contemporary Cinema, 2020, no. 8. In Chinese.

[6] Martin Heidegger. “The Age of the World Picture.” In *The Question Concerning Technology and Other Essays*, translated by William Lovitt, pp. 115-154. New York: Harper & Row, 1977.

[7] Sean Cubitt. *Finite Media: Environmental Implications of Digital Technologies*. Duke University Press, 2017, p. 3.

[8] Tao Yujie. “World, Play, and the Planetary: An Analysis of Three Core Concepts in Axelos's Philosophy and Their Interrelations.” French Philosophy Studies, 2023, no. 1. In Chinese.

[9] Han Xiaoqiang. “Toward Planetary Cinema.” Research on Art Communication, 2024, no. 2. In Chinese.

[10] Andrea Wulf. *The Invention of Nature*. Translated into Chinese by Bian He. Hangzhou: Zhejiang People's Publishing House, 2018, pp. 12-13. Chinese edition.

[11] Hannah Arendt. “Man's Conquest of Space.” The American Scholar 32, no. 4 (1963): 527-540.

[12] Gayatri Chakravorty Spivak. *Death of a Discipline*. Translated into Chinese by Zhang Xu. Beijing: Peking University Press, 2014, p. 90. Chinese edition.

[13] OpenAI. “Video Generation Models as World Simulators.” February 15, 2024. https://openai.com/index/video-generation-models-as-world-simulators/

[14] Han Xiaoqiang. “Toward Planetary Cinema.” Research on Art Communication, 2024, no. 2. In Chinese.

[15] Sasha Litvintseva. *Geological Filmmaking*. London: Open Humanities Press, 2022, p. 88.

[16] Elizabeth A. Povinelli. *Geontologies: A Requiem to Late Liberalism*. Duke University Press, 2016, p. 40. The source lists London as the place of publication.

[17] Roberto Casati and Achille C. Varzi. *Holes and Other Superficialities*. Cambridge, MA: MIT Press, 1994, p. 11.

[18] Wu Bingcheng. “Body Image, Identity Dissociation, and Bionic Devices: Research on VR Imagery Based on the Rubber Hand Illusion.” Film and Television Production, 2025, no. 11. In Chinese.

[19] Benjamin H. Bratton. *The Stack: On Software and Sovereignty*. Cambridge, MA, and London: MIT Press, 2016, p. xviii.

[20] Kate Crawford. *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Translated into Chinese by Ding Ning, Li Hongcheng, and Fang Wei, under the Chinese title Beyond Technology: Artificial Intelligence in Social Connections. Beijing: China Science and Technology Press, 2024, p. 29. Chinese edition.

[21] Yuk Hui. “For a Planetary Thinking.” e-flux journal 114, December 2020. https://www.e-flux.com/journal/114/366703/for-a-planetary-thinking/

[22] Microsoft. "*Microsoft Flight Simulator*: The Future of Game Development." Microsoft Developer, July 15, 2021. https://developer.microsoft.com/en-us/games/articles/2021/07/microsoft-flight-simulator-the-future-of-game-development/

[23] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, et al. "Generative Agents: Interactive Simulacra of Human Behavior." arXiv, April 7, 2023. https://arxiv.org/abs/2304.03442

[24] Kate Crawford and Vladan Joler. "Anatomy of an AI System." September 7, 2018. https://anatomyof.ai/

[25] International Energy Agency. "Energy and AI: Executive Summary." April 10, 2025. https://www.iea.org/reports/energy-and-ai/executive-summary

Translation Notes on Terminology and Attribution

These notes distinguish translation choices and bibliographic checks from the author's argument.

Planetarity and planetarization. "Planetarity" translates the theoretical concept of the planetary as irreducible to a controllable globe. "Planetarization" denotes the process of planetary technological mobilization. "Planetary thinking" is retained for Yuk Hui's formulation; "noodiversity" and "technodiversity" follow his English terminology.

World picture, worldbuilding, and worldmaking. "World picture" follows the established English title of Heidegger's essay. "Worldbuilding" generally renders the construction of worlds; "worldmaking" emphasizes the generative production of a world. "Narrative autonomy" comprises generative, causal, and normative autonomy, rather than simply the replacement of a human author by an automated system.

Names and titles. Standard forms include André Bazin, Kostas Axelos, Gayatri Chakravorty Spivak, Sasha Litvintseva, Elizabeth A. Povinelli, Pierre Teilhard de Chardin, Benjamin H. Bratton, Yuk Hui, Kate Crawford, and Vladan Joler. The body uses Povinelli's name without reproducing the source's specific professional designation. Crawford's book title is standardized to *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*; the cited Chinese edition and page are preserved in note 20.

Unresolved source attribution. The supplied text identifies the Dreamix team in note 2 with Google. This affiliation and the claim that its article first introduced AIGW could not be independently confirmed; the translation marks the affiliation as the source text's identification. This should not be conflated with verification of a priority claim.

Verification sources. Terminology and names were checked against the following primary sources and publisher records: Yuk Hui, "For a Planetary Thinking" (https://www.e-flux.com/journal/114/366703/for-a-planetary-thinking/); Columbia University Press, *Death of a Discipline* (https://cup.columbia.edu/book/death-of-a-discipline/9780231556873/); Meson Press, *Introduction to a Future Way of Thought* (https://meson.press/books/introduction-to-a-future-way-of-thought/); Open Humanities Press, *Geological Filmmaking* (https://www.openhumanitiespress.org/books/titles/geological-filmmaking/); Duke University Press, Geontologies (https://www.dukeupress.edu/geontologies); and Kate Crawford's academic bibliography (https://katecrawford.net/pubs). Technical references were also checked against the original Genie and Generative Agents papers, Microsoft Research's Muse/WHAM project, the Roblox developer announcement, and the IEA report cited in the body.